



Customer Retention Anaylsis

Objective :

The main aim of this project is to uncover insights into Flipkart's customer service operations and identify key factors affecting customer retention. By analysing 30,000 customer service call records, this project aims to understand what drives customer satisfaction scores and recommend actionable improvements.

Defining Key Metrics :

CSAT Score (csat_score): A measure of customer satisfaction with the support they received, calculated from feedback ratings after interactions. In this dataset, CSAT is rated on a scale of 1 to 10. The overall average CSAT score across all 30,000 calls is 5.54.

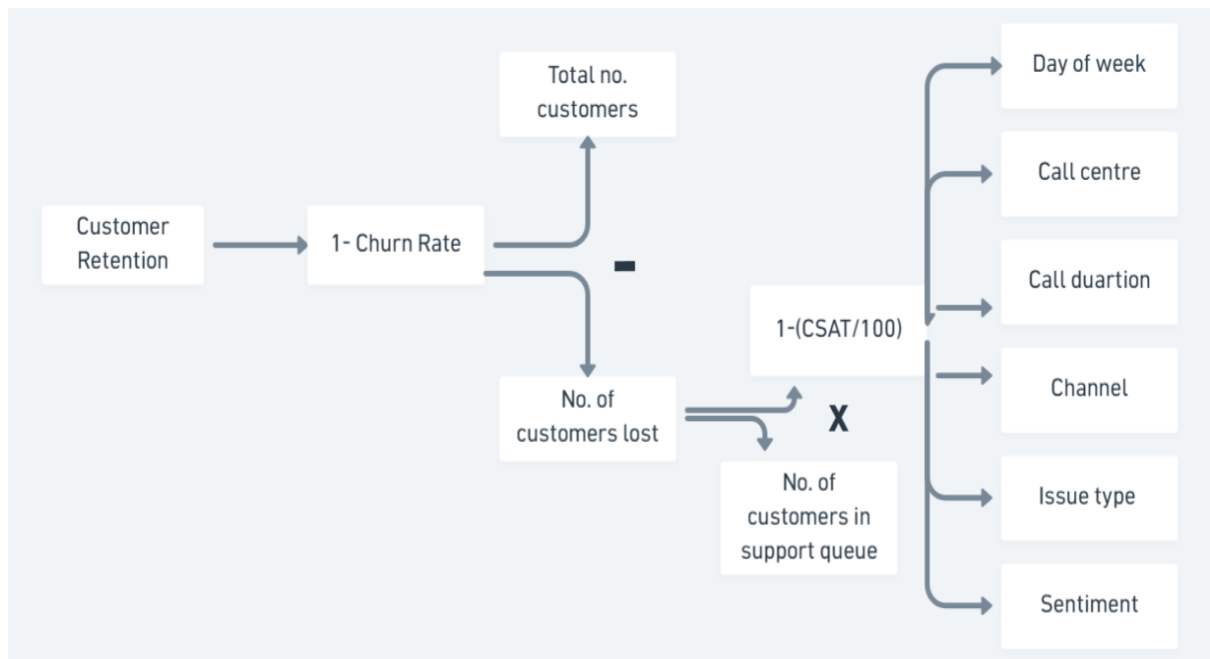
Sentiment: Customer attitude during a service interaction, categorised as Very Positive, Positive, Neutral, Negative, or Very Negative based on their feedback during the call.

Call Duration: The length of each customer service call in minutes, used to understand whether longer calls affect satisfaction.

Support Channel: The method through which the customer contacted support - Call-Center, Chatbot, Email, or Web.

Day of Week: The day on which the call was made, derived from the call timestamp to identify patterns across the week.

METRIC TREE :



Data Cleaning:

Data cleaning was performed to ensure the quality and reliability of the analysis. The following actions were taken:

Data Import: The CSV file containing 30,000 rows of customer service call data was imported into Excel and saved as an .xlsx file.

Handling Missing Values: CSAT scores were missing for a large portion of rows. These were retained rather than deleted as they still contribute to call volume analysis. Missing values in city and state columns (marked as #N/A) were removed using Find and Replace.

Standardising Date Formats: The call_timestamp column contained two mixed formats - MM/DD/YYYY and MM-DD-YYYY. This was fixed using the Text to Columns feature in Excel, standardising all dates into a consistent format.

New Columns Added:

DAY - Derived from the call timestamp to show the day of the week for each call. Formula: =TEXT(F2,"DDDD")

CORRECTED DATE - Extracted the clean date from the timestamp. Formula: =TEXT(F2,"DD-MM-YYYY")

HELPING COLUMN - Created to assist with date standardisation during cleaning.

HYPOTHESIS

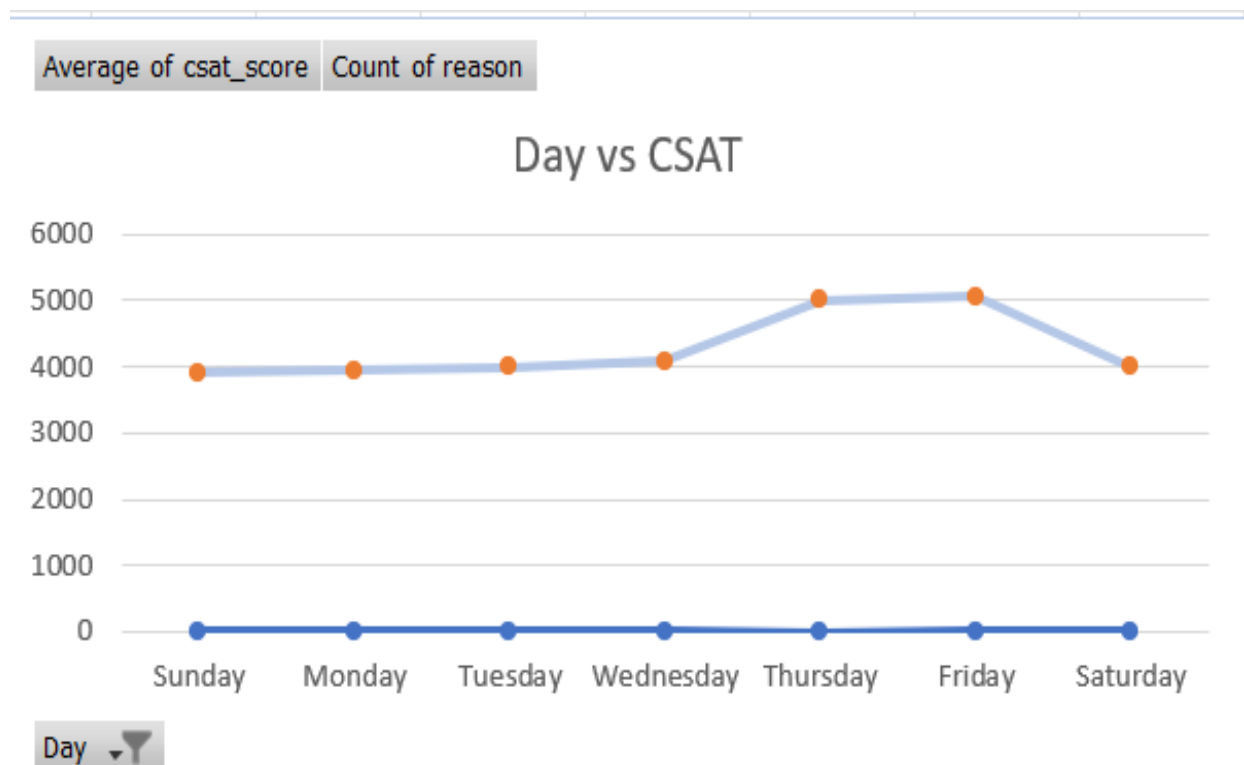
Hypothesis 1: The day of the week with higher issue volumes influences CSAT scores

Data columns used: csat_score, DAY, reason (Count)

Analysis: Thursday had the highest call volume at 5,001 calls and the lowest average CSAT of 5.43. Wednesday performed best with a CSAT of 5.61 despite moderate volumes of 4,077 calls.

Conclusion: Higher issue volumes on Thursdays negatively impact CSAT scores, likely due to increased agent workload reducing service quality. Wednesdays show the best performance.

Recommendation: Allocate additional staffing on Thursdays and Fridays to handle peak volumes without compromising service quality.



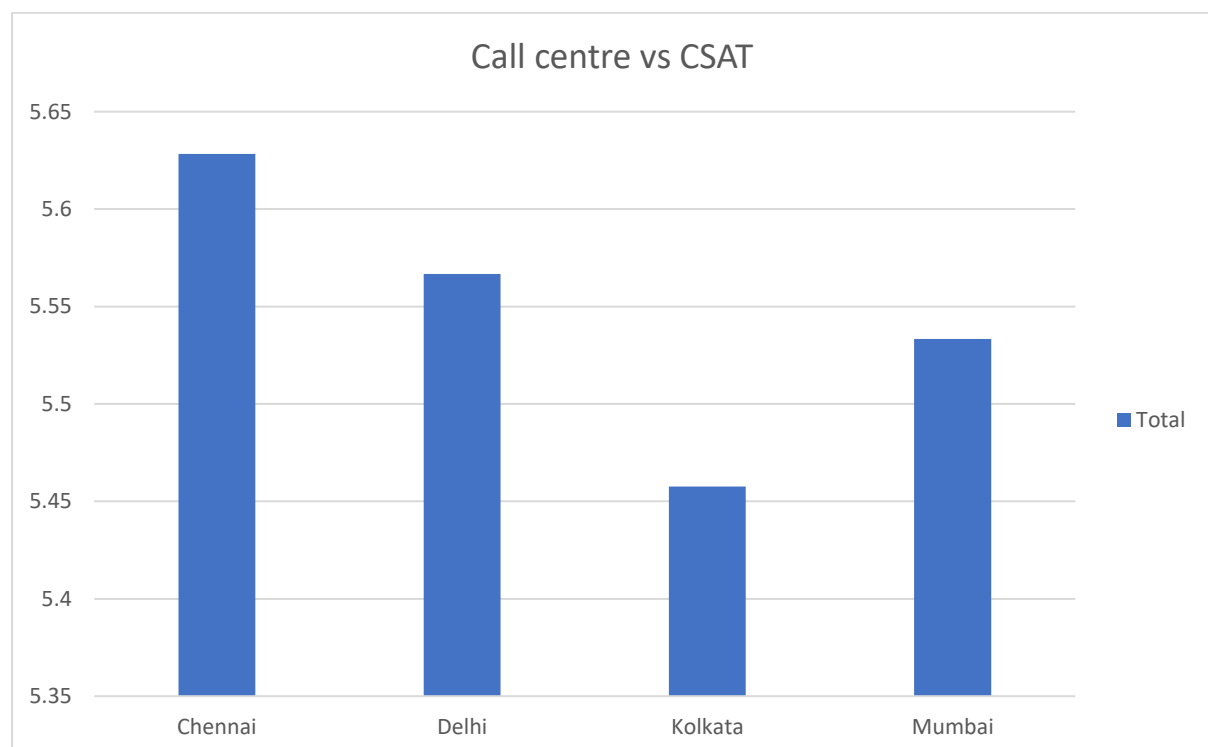
Hypothesis 2: Certain call centers have consistently lower CSAT scores

Data columns used: csat_score, call_center

Analysis: Kolkata had the lowest average CSAT of 5.46 while Chennai had the highest at 5.63. The overall average CSAT across all centers is 5.54.

Conclusion: Kolkata consistently underperforms all other call centers. This suggests quality issues specific to that center rather than system-wide problems.

Recommendation: Investigate recurring issues in Kolkata. Implement targeted training focused on communication and issue resolution skills.



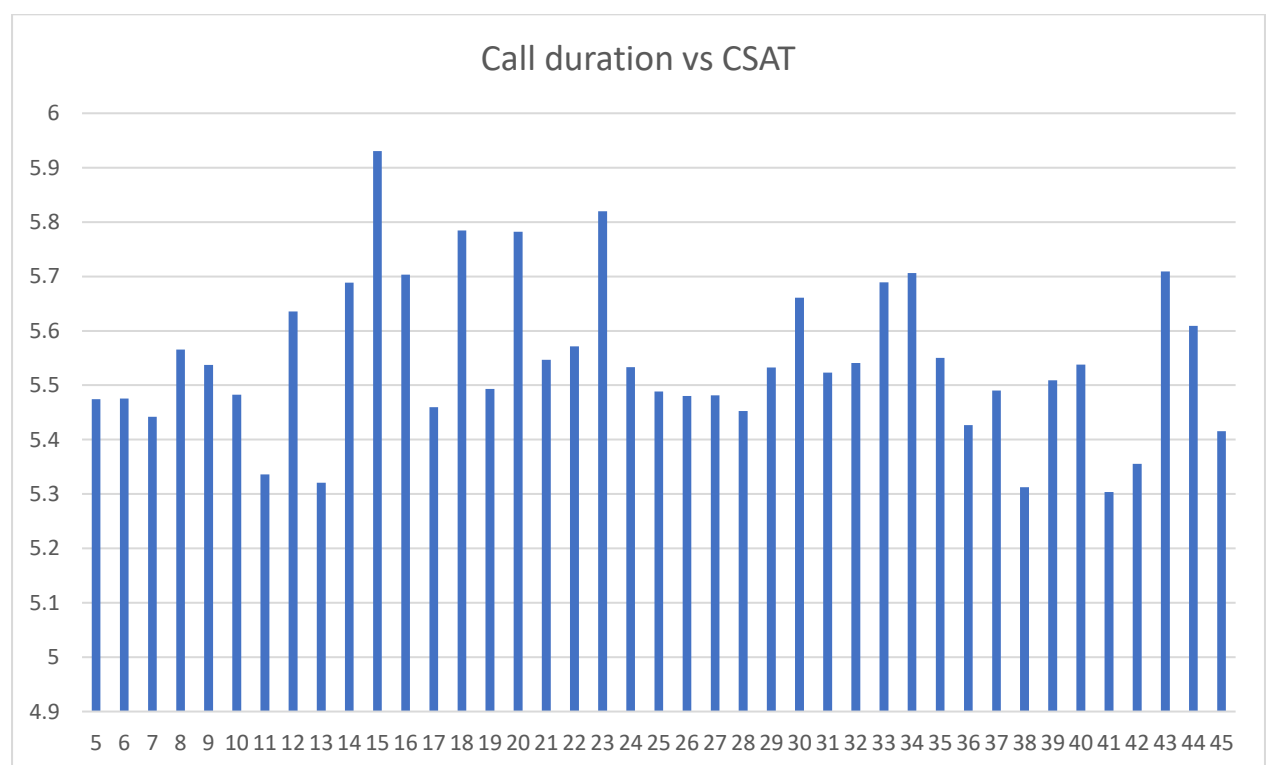
Hypothesis 3: Call duration influences CSAT scores

Data columns used: csat_score, call duration in minutes

Analysis: Call duration ranged from 5 to 45 minutes. Shorter calls under 20 minutes had slightly higher average CSAT scores compared to longer calls of 40 minutes and above. For example, calls lasting 15 minutes scored 5.93 while calls at 45 minutes dropped to 5.42. Calls at 38 minutes scored 5.31 and calls at 41 minutes scored 5.30 are among the lowest in the dataset.

Conclusion: Longer calls may indicate unresolved issues, leading to lower satisfaction. When a customer stays on a call for a long time, it often means their problem is not being solved efficiently.

Recommendation: Train agents to resolve issues efficiently in shorter call times. Introduce call monitoring to flag calls exceeding 35 minutes for supervisor review.



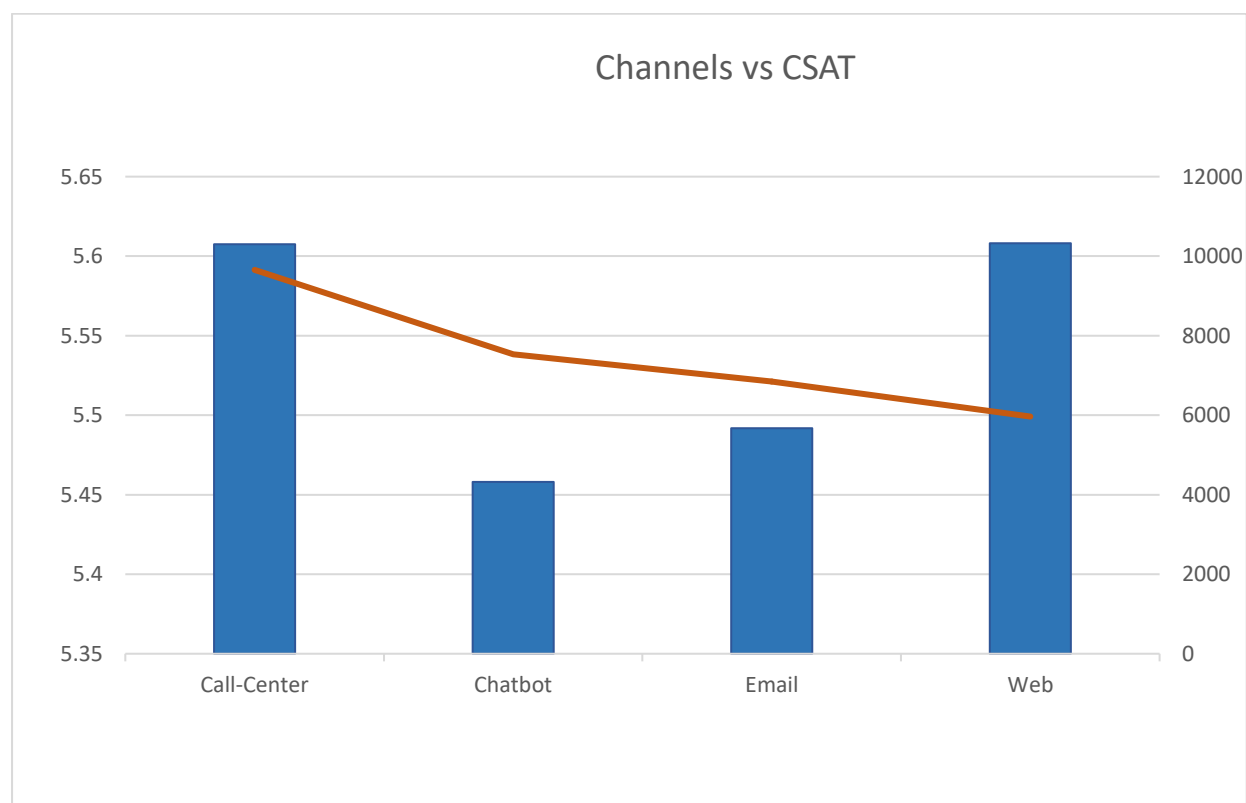
Hypothesis 4: Support channel influences CSAT scores

Data columns used: csat_score, channel, id (count)

Analysis: Web and Call-Center both scored 5.61. Chatbot had the lowest CSAT of 5.46 despite handling 7,533 calls ,
the second highest volume after Call-Center.

Conclusion: Chatbot consistently underperforms human-handled channels. Customers clearly prefer human interaction for issue resolution.

Recommendation: Improve chatbot accuracy and add a clear escalation option to transfer unresolved queries to a human agent immediately.



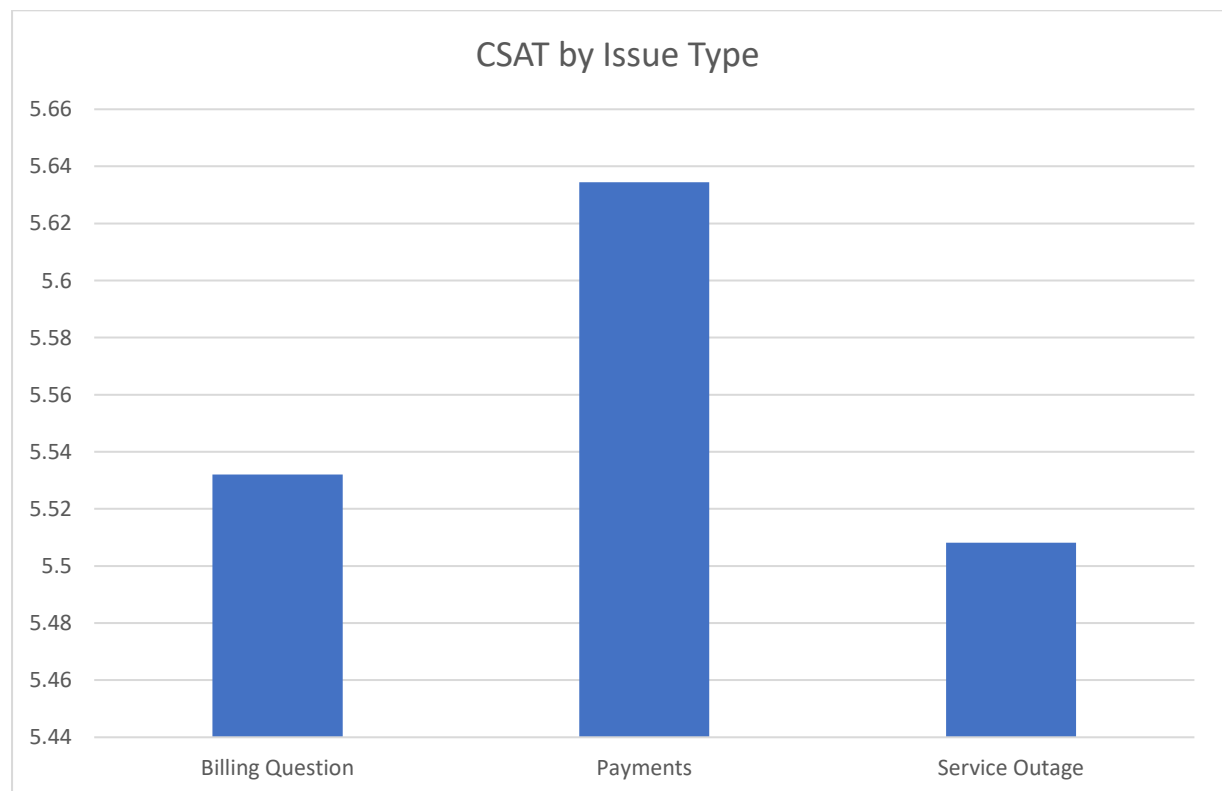
Hypothesis 5: Certain issue types are associated with lower CSAT scores

Data columns used: csat_score, reason

Analysis: Customers reporting Service Outage issues recorded the lowest average CSAT score of 5.51, while Payments related issues achieved the highest at 5.63.

Conclusion: Service Outage issues have the greatest negative impact on customer satisfaction. Customers are clearly less tolerant of service disruptions compared to billing or payment concerns.

Recommendation: Create a dedicated fast-response team for Service Outage calls. Proactively notify customers during outages to reduce frustration and call volume.



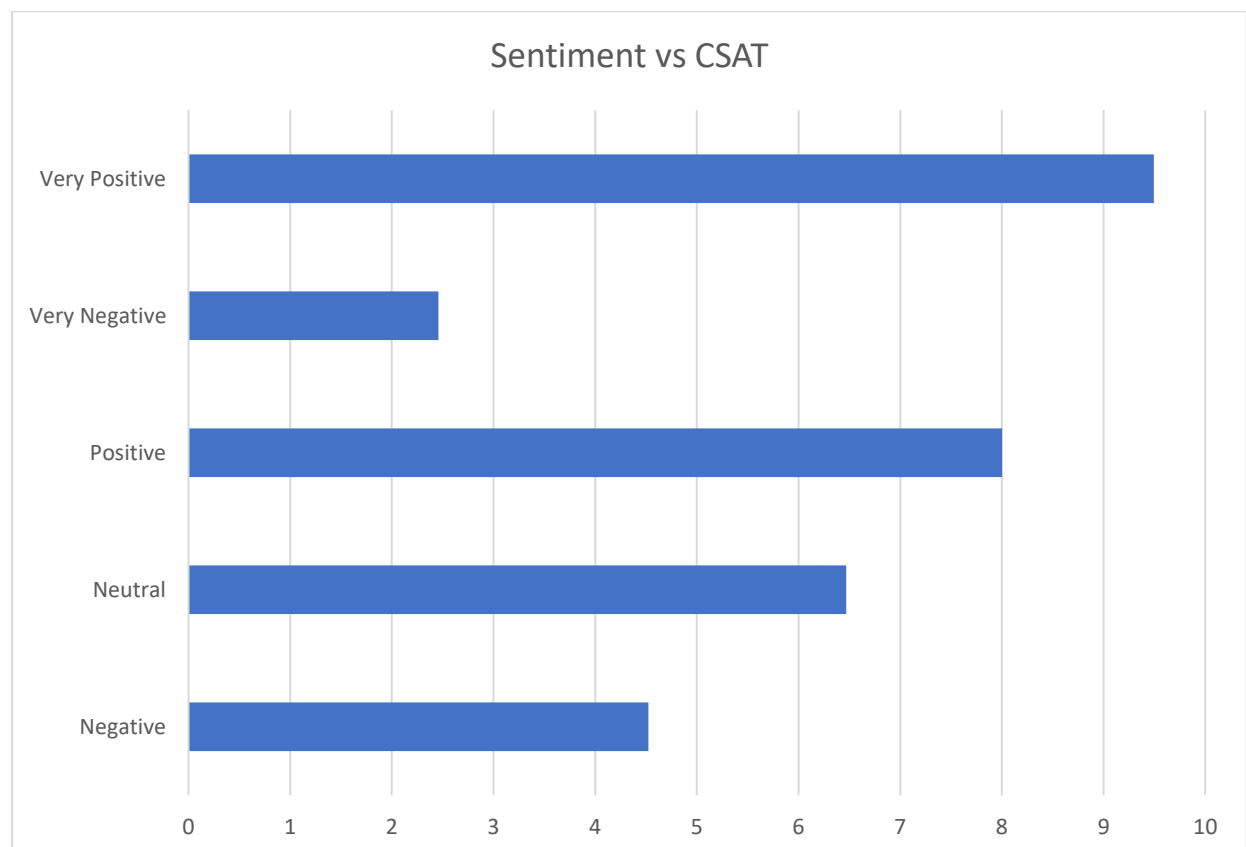
Hypothesis 6: Customer sentiment influences CSAT scores

Data columns used: csat_score, sentiment

Analysis: Very Negative sentiment calls scored only 2.46 while Very Positive calls scored 9.49, a gap of 7.03 points. This is the largest difference found in any hypothesis in the entire analysis.

Conclusion: Sentiment is the single strongest driver of CSAT scores. A customer who feels positive during a call will almost always give a high satisfaction score, while a negative customer will score very low.

Recommendation: Follow up personally on all Very Negative interactions. Invest in agent communication training to turn neutral and negative interactions into positive ones, as this will have the biggest direct impact on CSAT and customer retention.



Summary of Key Findings :

Based on the analysis of 30,000 Flipkart customer service calls with an overall average CSAT of 5.54 and average call duration of 25 minutes:

1. **Sentiment** is the strongest driver of CSAT - Very Negative calls score 2.46 while Very Positive calls score 9.49, a gap of 7 points. This is the most impactful finding in the entire project.
2. **Kolkata** call center consistently scores the lowest CSAT at 5.46 across all categories, suggesting quality issues specific to that center.
3. **Chatbot** is the weakest support channel with an average CSAT of 5.46, scoring lower than all human-handled channels including Call-Center (5.61), Web (5.61) and Email (5.49).
4. **Thursday** has the highest call volume at 5,001 calls and the lowest CSAT of 5.43, suggesting peak-day pressure negatively impacts service quality.
5. **Service Outage** issues are associated with the lowest CSAT scores at 5.51, while Payments issues score highest at 5.63.
6. **Longer calls** of 40+ minutes tend to have lower CSAT scores than shorter calls under 20 minutes, suggesting unresolved issues lead to longer and less satisfying interactions.

Recommendations :

- Improve chatbot responses and provide an easy option to connect with a human agent when needed.
- Increase support staff on Thursdays and Fridays to handle higher call volumes efficiently.
- Provide additional training for agents at the Kolkata call center to improve customer interactions.
- Focus on resolving customer issues faster to avoid long call durations and improve satisfaction.
- Create a dedicated team to handle service outage complaints and provide timely updates to customers.
- Follow up with customers who report negative experiences to better understand and address their concerns.
- Regularly monitor CSAT scores across channels, call centers, and issue types to identify areas for improvement.
- Encourage agents to use positive and empathetic communication, as customer sentiment has the strongest impact on satisfaction.